Plan of Attack

Costco Australia / Synthetic Control

Research question: Does the entry of a Costco gas station into an Australian local market cause nearby competitor stations to lower their retail fuel prices?

Section 1, Raw Data Description

We use four datasets, described separately below: three publicly accessible Australian state fuel-price registries that we combine into a unified station-month panel, plus a hand-collected list of Costco fuel-station opening dates. The three registries together yield approximately 17.4 million station-day price records across 6,084 unique stations between January 2018 and April 2026.

1.1 NSW FuelCheck

NSW FuelCheck is maintained by the New South Wales Government and has required mandatory price reporting from every retail fuel station in NSW (and, since November 2023, the ACT) since August 2016. Historical records are published as monthly Excel files via data.nsw.gov.au; we pulled 93 monthly files covering December 2016 through January 2026, totalling **4,401,971 price-update records** with 8 columns. Each row is one price-update event: a station × fuel type × the timestamp at which the operator changed (or re-confirmed) its posted price. FuelCheck stores price-change events rather than daily snapshots. Composite unique key: `(ServiceStationName, Address, FuelCode, PriceUpdatedDate)`. The file contains 4,813 unique stations (4,543 after the name normalization used in the live-API coordinate join, see Section 2(f)), 62 unique brands, 565 unique postcodes, and 11 unique fuel-type codes (regular unleaded ULP/U91/E10, premium PUL95/PUL98, diesel DSL/PDL, LPG, and others). Coordinates are not in the historical files; we attached lat/lng by joining station name + postcode to a one-time pull from the live FuelCheck API.

Field	Description
`ServiceStationName`	Station name (operator-provided)
`Address`	Street address
`Suburb`	Suburb
`Postcode`	Postcode (4-digit)
`Brand`	Brand affiliation (Costco, Caltex, Shell, BP, etc.)
`FuelCode`	Fuel-type code (U91, E10, PUL95, PUL98, DSL, LPG, ...)
`PriceUpdatedDate`	Timestamp at which price was reported
`Price`	Posted retail price (cents per liter)

1.2 Queensland Fuel Price Reporting Scheme

The Queensland Fuel Price Reporting Scheme is administered by Queensland Treasury; mandatory station reporting began in December 2018. Historical records are published as monthly CSVs via data.qld.gov.au; we pulled 85 monthly files covering December 2018 through December 2025, totalling

4,970,650 price-change records with 13 columns. Each row is one price-change event: a station × fuel type × the UTC timestamp at which the price changed. Composite unique key: `(SiteId, Fuel_Type, TransactionDateutc)`. The file contains 1,883 unique stations, 37 brands, 375 postcodes, and 10 fuel-type strings. Coordinates are present inline.

Field	Description
`SiteId`	Stable per-station numeric identifier
`Site_Name`	Station name
`Site_Brand`	Brand affiliation
`Sites_Address_Line_1`	Street address
`Site_Suburb`	Suburb
`Site_Post_Code`	Postcode (4-digit)
`Site_Latitude / Site_Longitude`	WGS84 coordinates (used for great-circle distance to Costco)
`Fuel_Type`	Fuel-type label (Unleaded 91, e10, Premium 95, ...)
`Price`	Posted retail price in tenths of a cent (1899 = 189.9 ¢/L)
`TransactionDateutc`	UTC timestamp of the price change

1.3 WA FuelWatch

WA FuelWatch is maintained by the Western Australia Government and has required daily mandatory price reporting from every retail fuel station in WA since January 2001, the longest fuel-price history of any Australian state. Historical records are published as monthly CSVs via data.wa.gov.au; we pulled 100 monthly files covering January 2018 through April 2026, totalling **8,047,153 daily-snapshot records** with 11 columns. (We chose not to pull the full 2001–2017 archive because no usable Costco event has a pre-period that benefits from extending earlier than 2018.) Each row is a daily snapshot (a station × fuel type × calendar day), and unlike NSW and QLD, FuelWatch publishes one record per station per day regardless of whether the price changed. Composite unique key: `(TRADING_NAME, ADDRESS, PRODUCT_DESCRIPTION, PUBLISH_DATE)`. The file contains 1,423 unique stations, 47 brands, 200 postcodes, and 7 fuel-type strings. Coordinates are not in the historical files; we attached lat/lng by joining station name + postcode to a one-time pull from the live FuelWatch API.

Field	Description
`PUBLISH_DATE`	Calendar day of the snapshot
`TRADING_NAME`	Station name
`BRAND_DESCRIPTION`	Brand affiliation
`PRODUCT_DESCRIPTION`	Fuel-type label (ULP, 98 RON, Diesel, LPG, ...)
`PRODUCT_PRICE`	Posted retail price (cents per liter)
`ADDRESS`	Street address

`LOCATION`	Suburb / locality
`POSTCODE`	Postcode (4-digit)
`AREA_DESCRIPTION / REGION_DESCRIPTION`	Government-defined regional grouping (used as a secondary geographic check)

1.4 Costco gas station opening dates (hand-collected)

We hand-compiled a 10-row table of every Costco fuel station in Australia, recording each station's state, suburb, postcode, coordinates, and treatment date. Each treatment date is the station's first-observed date in the relevant state's price registry, then cross-validated against announced fuel-station opening dates and trade press (Section 2(d)). The composite unique key is `name`. Of the 10 stations, four are usable as treated units in the synthetic-control analysis (Coomera, Casuarina, Perth Airport, Lake Macquarie); the remaining six are excluded from formal estimation but appear as descriptive context.

Outcome and treatment construction

Our main outcome is the monthly mean unleaded retail price (cents per litre), constructed in Section 2 by filtering to regular unleaded (FuelCode in {U91, E10, P91} for NSW; Fuel_Type in {Unleaded 91, e10} for QLD; PRODUCT_DESCRIPTION = ULP for WA) and aggregating in two stages: (station × month), then across stations within the unit. The unit differs by group: each *treated* unit is a (Costco × month) cell averaging non-Costco stations within 5 km of one of the four treated Costco coordinates; each *donor* unit is a (postcode × month) cell averaging non-Costco stations within the postcode. The treatment indicator (a per-Costco binary post-opening flag) is constructed in Section 2 by merging the hand-collected treatment-date table onto the treated panel. The donor pool consists of postcodes located more than 20 km from any Costco that also clear the downstream coverage filters (Section 2(g)).

Summary statistics, postcode-month analysis sample (3 states, Jan 2018 – Apr 2026)

Variable	n	Mean	SD	Min	p25	Median	p75	Max
Main outcome								
Mean unleaded price (¢/L)	17,953	160.5	28.4	92.0	137.3	162.2	183.6	259.0
Main covariates								
Station distance to nearest Costco (km)	5,980	204.8	357.4	0.0	14.8	48.3	216.7	2,214

Counts: 84 stations classified as treated (within 5 km of a treated Costco), 3,926 donor-eligible (>20 km from every Costco), 1,970 in the 5–20 km exclusion donut. The 17,953 postcode-month rows above are the donor candidate panel before geographic and quality filters; the post-filter SC donor pool is 17,578 cells (Section 2(g)).

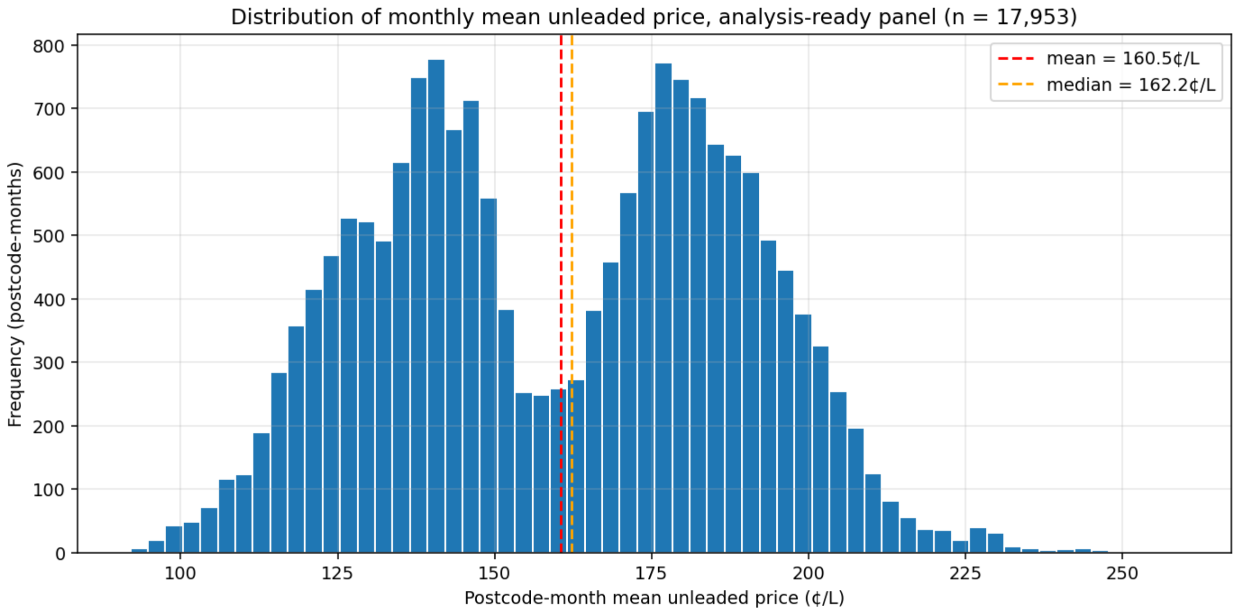


Figure 1. *Distribution of postcode-month mean unleaded prices across the 17,953 panel observations. The distribution is approximately bimodal: the lower mode reflects the 2018–2020 price regime, the higher mode reflects the post-2022 elevated price era. Mean = 160.5 ¢/L, median 162.2 ¢/L. No observations outside 92–259 ¢/L after our 50–500 ¢/L sanity filter.*

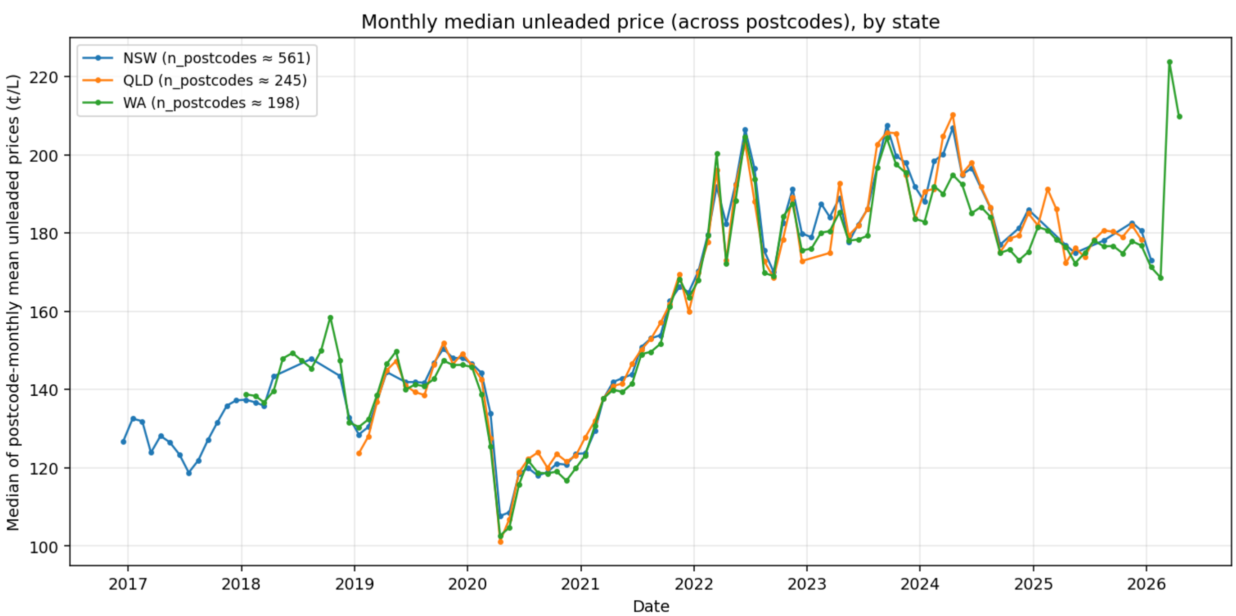


Figure 2. *Median of postcode-monthly mean unleaded prices over time, by state, January 2018 to April 2026. All three states track each other closely throughout: prices fall in 2018–2020 (peaking with COVID demand collapse in early 2020), rise sharply through 2022 (Russia/Ukraine energy shock), dip during the Mar–Sep 2022 fuel-excise cut, and remain elevated through 2026. State-level co-movement supports our use of within-state donor pools to absorb common national shocks.*

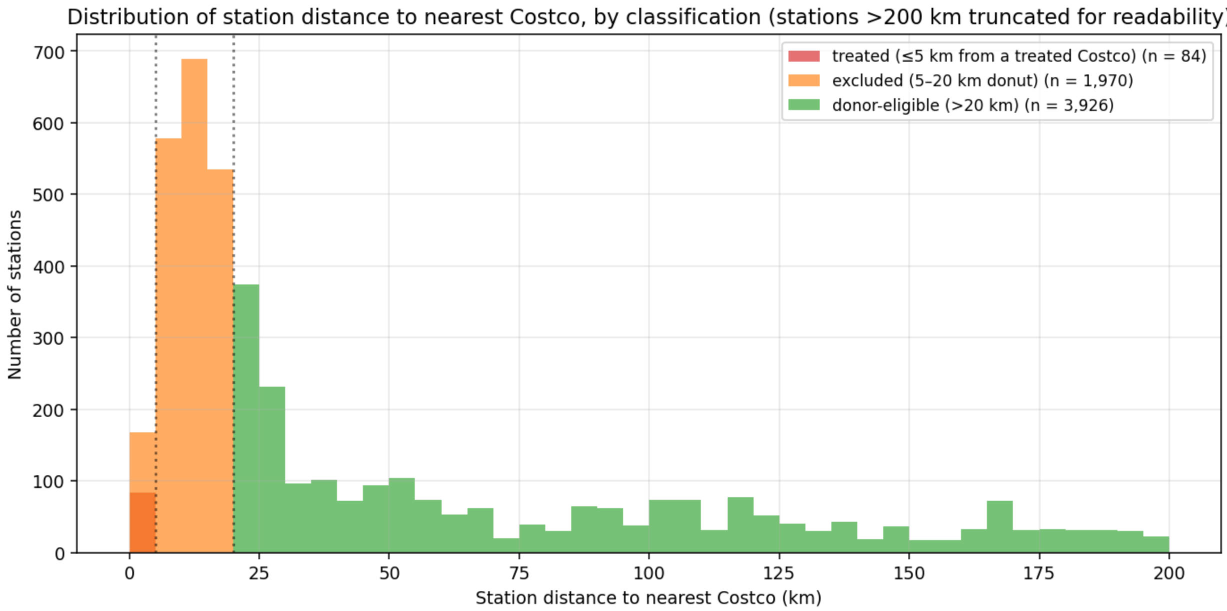


Figure 3. *Distribution of station distance to the nearest Costco, split by classification. Vertical dotted lines mark the 5 km treated boundary and the 20 km donor-pool boundary; stations 5–20 km from any Costco are excluded from both groups to avoid spillover contamination. The bulk of donor-eligible stations sit well outside any plausible Costco competitive footprint, supporting the donor pool's status as an untreated counterfactual. Stations beyond 200 km are truncated for readability.*

Australian Costco fuel stations by treatment status

Costco	State	Treatment date	Pre months	Post months	Status
Coomera	QLD	May 2023	51	31	Treated
Casuarina	WA	Nov 2022	58	42	Treated
Perth Airport	WA	Feb 2020	26	74	Treated
Lake Macquarie	NSW	May 2022	58	35	Treated
Marsden Park	NSW	Jul 2017	11	105	Excluded (pre too short)
Auburn	NSW	Aug 2025	107	8	Excluded (post too short)
Canberra Airport	ACT	Nov 2023	0	29	Excluded (no pre)
Casula	NSW	Dec 2016	4	113	Excluded (pre-registry)
North Lakes	QLD	Jan 2019	1	89	Excluded (pre-registry)
Ipswich	QLD	Mar 2019	3	87	Excluded (pre-registry)

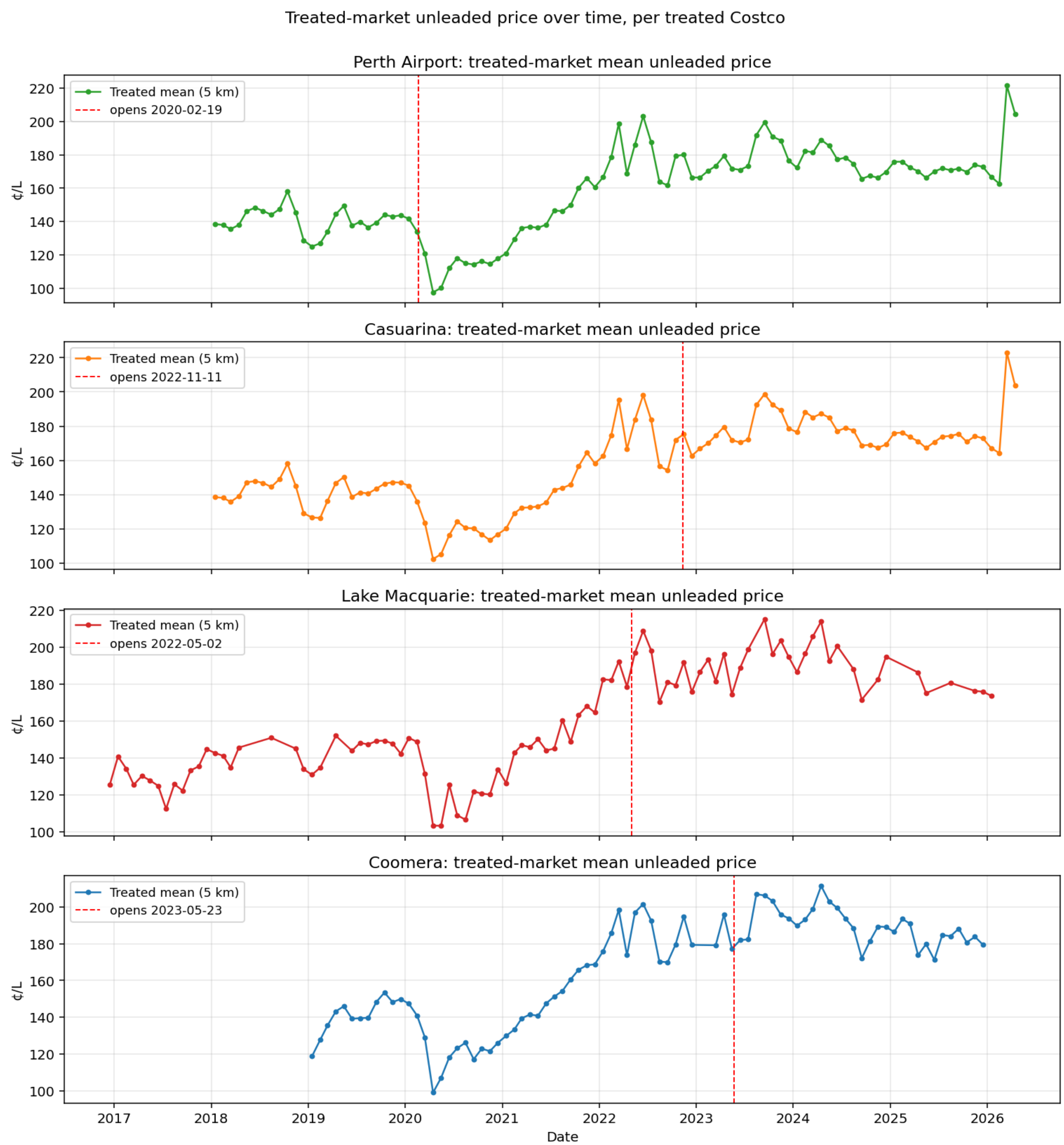


Figure 4. *Treated-market mean unleaded price over time for each of the four treated Costcos, with the vertical red dashed line marking the validated treatment date (Section 2(d)). Eyeballing pre-period trajectories before fitting any model surfaces data-quality issues and high-leverage points: Casuarina's sparse treated-station count (~3 stations) shows up as month-to-month volatility that the other three Costcos don't exhibit; Coomera's registry coverage gradually builds up through 2018–2020 before stabilizing.*

Data quality issues

Price units differ across registries. NSW prices are in cents/L (e.g. `158.7`), QLD prices are in tenths of cents (`1899` = 189.9 ¢/L), WA prices are in cents/L. Each registry's ingest applies the appropriate scaling so the panel is uniform in cents/L.

Coordinates missing in historical NSW and WA files. NSW FuelCheck XLSX records carry only address text; WA FuelWatch records carry address and postcode (no lat/lng). QLD historical records do include coordinates inline. We resolved this by pulling the live NSW FuelCheck API (`/v1/fuel/prices`) and the live WA FuelWatch API (`/api/sites`) once and joining historical records to the live coords by (normalized_name, postcode). The live-API snapshot contains NSW 3,281 stations, WA 921 stations, QLD 1,882 stations, totalling 6,084. Stations that closed, rebranded, or were re-numbered before April 2026 are absent from the snapshot, so historical rows for them fail the join: the snapshot misses 2,376 of 4,543 NSW historical stations (52.3%) and 673 of 1,423 WA historical stations (47.3%); QLD is unaffected because its historical files carry coordinates inline. The loss falls almost entirely on the donor pool: all four treated Costcos join cleanly. Row-level impact on the analysis panel is in Section 2(g).

Brand renaming in NSW (May 2022). Every NSW Costco station was renamed in the registry from e.g. "Costco Marsden Park" to "Costco Marsden Park (Members only)" on 2 May 2022. We strip the "(Members only)" suffix before matching so the rename is invisible to the analysis pipeline.

Outliers in `price`. Quotes below 80 ¢/L or above 280 ¢/L are implausible for Australian retail unleaded post-2018 (national averages hover 130–220 ¢/L). After our 50–500 ¢/L sanity filter, the post-aggregation panel of 17,953 postcode-months has no observations outside 92–259 ¢/L, well within plausible Australian range.

Treated-Costco coverage variance. Coomera's pre-2020 QLD coverage is sparse because the QLD scheme began in December 2018 and Coomera-area stations entered the registry gradually; the relevant pre-period (Sep 2020 → May 2023) is fully covered. Casuarina has only ~3 unique stations within 5 km contributing to the treated mean, reflecting a sparse coastal residential market; coverage is 100 % but the treated mean is noisier than for the other three Costcos. Lake Macquarie has a small number of intermittent monthly gaps (largest 3 months) that do not impair pre-period fitting.

Donor pool construction. Out of all postcodes with any non-Costco unleaded data, the donor pool drops 362 postcodes within 20 km of any Costco (potential contamination from Costco's competitive footprint), 446 with fewer than 3 stations on average per month (means too noisy), and 0 with fewer than 24 months of observations. **196 donor postcodes** survive all filters: NSW 104, QLD 56, WA 36, totalling 17,578 (postcode × month) panel observations. Per-state breakdown of the 362 / 446 / 0 drops is in Section 2(g).

Multiple records per station-day. NSW and QLD both record price-update events rather than daily snapshots, so a station appears multiple times within a day if it changed prices on multiple fuel types. These are not duplicates; they are legitimate event-level records and are correctly aggregated to monthly means by the panel build.

Contingencies

If histograms of postcode-month unleaded prices reveal extreme outliers in any donor postcode (e.g. a one-time data-entry anomaly from a single station dominating a small postcode), the panel is winsorised at the 1st and 99th percentile before synthetic-control weight fitting. Postcode-months with fewer than 3 contributing observations after the panel build are flagged as missing, not retained at face value. If pre-period fit quality for any treated Costco is demonstrably poor (Section 3, visible mismatch between treated and synthetic in the pre-period), that Costco is presented with a documented limitation, not as a headline result; Casuarina is the most likely candidate given its small treated-station count.

Plan of Attack: Section 2, Sample Construction Plan

(a) Sample window

Endpoints. The pre-period for each treated Costco runs from the start of its state's price registry (December 2016 for NSW, January 2018 for WA, December 2018 for QLD) up to the day before that Costco's validated treatment date (see (d)). The post-period runs from the treatment date through April 2026, the most recent registry pull. Per-Costco pre-period lengths range from 26 (Perth Airport) to 58 months (Casuarina, Lake Macquarie); post-period lengths from 31 (Coomera) to 74 months (Perth Airport).

Pre-period length. Synthetic control fits its weights on the entire pre-period *trajectory* of the dependent variable, not just on a level. More pre-period observations tighten identification rather than dilute it, the opposite of the DID logic in which long pre-periods can introduce structural breaks into the parallel-trends assumption. Same-state donors experience the same world-shifts (oil-price regime changes, vehicle fleet turnover, federal excise adjustments) over the same window, so common shocks are absorbed into the synthetic by construction. We use as much pre-period as the data allow.

COVID. The 2020 demand collapse and recovery is the only structural break large enough to warrant explicit handling. Three of the four treated Costcos opened well after the COVID disruption (Lake Macquarie May 2022, Casuarina November 2022, Coomera May 2023); for those three the COVID period sits entirely inside the pre-period and is matched by the synthetic by construction. **Perth Airport (treated 19 February 2020)** opens approximately four weeks before Australia's COVID demand collapse begins; this is the most fragile case in the analysis. Its pre-period is entirely pre-COVID, but its early post-treatment months are dominated by pandemic-driven price movements rather than competitive response to Costco's entry. The synthetic counterfactual is built from WA donor postcodes that experienced the same pandemic shock, so the post-treatment gap nets out COVID provided the pre-period weights track the treated trajectory through January 2020. As a Section 3 robustness check, Perth Airport is re-estimated with March–December 2020 dropped from both panels.

(b) Other sample restrictions

Treated unit. For each treated Costco, the treated unit is the *local market* defined as all non-Costco retail stations within 5 km of the Costco's coordinates, summarised each month as the simple mean of those stations' monthly mean unleaded prices. We exclude Costco's own price quotes from the treated mean because the research question concerns competitor response, not Costco's own pricing.

Donor pool. Each donor unit is a 4-digit Australian postcode, summarised the same way (monthly mean of non-Costco unleaded price across all stations within the postcode). Postcodes are the natural geographic unit of the three state registries and are consistent across NSW, QLD, and WA. A postcode is excluded from the donor pool if any of its stations lie within 20 km of *any* Costco fuel station (treated, excluded, or otherwise), so that donor postcodes are unambiguously outside any plausible Costco competitive footprint.

Asymmetric radii (5 km treated, 20 km donor buffer). The economics literature on retail fuel-price competition (Houde 2012; Eckert 2013) finds price effects concentrated within roughly 1–5 miles. 5 km

(~3.1 mi) sits at the conservative end of that range, small enough to capture genuinely competing stations but not so small as to drop most of the local market. The 20 km donor buffer is intentionally a 4× multiple, providing a wide exclusion donut so that no donor postcode is partly affected by a Costco that isn't the one we're studying. The asymmetry (small treated radius, large donor buffer) biases the design *against* finding an effect: narrow treated definition shrinks the treated population to the most exposed stations, and the wide buffer shrinks the donor pool to truly untreated postcodes.

Outcome variable. Regular unleaded (FuelCode in {U91, E10, P91} for NSW; `Fuel_Type` in {Unleaded 91, e10} for QLD; `PRODUCT_DESCRIPTION` = ULP for WA). Unleaded is the highest-volume retail product in Australia, comparable across the three state schemes, and Costco's headline competitive product.

Donor-pool measurement-quality filters. A postcode is additionally excluded if it averages fewer than 3 unique stations per month (monthly means based on 1–2 stations are too noisy to serve as credible counterfactual building blocks) or has fewer than 24 months of any observed coverage (weights cannot be reliably fit on a short pre-period).

Treated-Costco inclusion rule. A Costco fuel station is treated as a synthetic-control unit if and only if its registry history contains both **at least 24 pre-treatment months** and **at least 6 post-treatment months** in the relevant state's price registry. Four of Australia's ten Costco fuel stations satisfy both conditions (Coomera, Casuarina, Perth Airport, Lake Macquarie); six do not, and appear as descriptive context in Section 1's coverage table. Exclusion reasons: Marsden Park (11 pre-months), Auburn (8 post-months), Canberra Airport (no pre; ACT joined NSW FuelCheck after this Costco opened), Casula and North Lakes (≤4 pre-months each; the Costco predates the relevant registry), Ipswich (3 pre-months, predates the QLD scheme).

Table 1. Threshold-and-defense summary.

Threshold	Value	Justification	Robustness check (Section 3)
Treated radius	≤5 km	Conservative end of IO-literature retail fuel-price spillover range (1–5 mi)	Re-run at 3 km and 8 km
Donor buffer	>20 km	4× the treated radius; keeps donors fully outside any Costco's competitive zone	Re-run at 15 km and 30 km
Donor stations / month	≥3	Below 3, monthly mean has high sampling variance	Re-run at ≥2 and ≥5
Donor coverage	≥24 months	Minimum pre-period length for stable SC weight fitting	Re-run at ≥36 months
Costco pre-period	≥24 months	Same logic, applied to treated	Not relaxed (would re-introduce excluded Costcos that fail the rule)
Costco post-period	≥6 months	Minimum to estimate a sustained-effect gap	Not relaxed
Plausible price range	50–500 ¢/L	Sentinel-value floor and ceiling; AU retail unleaded lives in 80–280	None; not an analytic threshold

(c) Dependent variable integrity

Construction. The dependent variable is the monthly mean unleaded retail price (cents per litre) at the treated 5 km ring or the donor postcode; see (e) for the two-stage construction. Section 1's Figure 1 shows the post-aggregation distribution (n = 17,953 postcode-months, range 92–259 ¢/L, approximately bimodal across the pre-2022 / post-2022 price-regime divide).

Sentinel-value filter. Individual price quotes outside 50–500 ¢/L are dropped before aggregation. This is a data-integrity filter, not winsorisation: it removes obvious registry artifacts (out-of-stock placeholder values like `9999` or `0.01`) rather than legitimate tail observations. Australian retail unleaded post-2018 has averaged 130–220 ¢/L; anything outside 50–500 is mechanical, not market.

Missing months. A postcode-month or treated ring-month with zero contributing station observations is left as missing (`NaN`), never imputed as zero, never carried forward. Synthetic-control fitting handles missing months by ignoring them in the pre-period objective; Section 1 reported that the largest run of missing months in any treated panel is three (Lake Macquarie), which is not material.

No winsorisation in the main specification. Two-stage monthly aggregation smooths individual-quote outliers: an extreme single quote at one station gets averaged with the rest of that station's month, and then with the rest of the ring or postcode. If a robustness check exposes a donor-postcode month driving an estimate, we winsorise the analysis panel at the 99th percentile and report both specifications side-by-side.

(d) Independent variable integrity

Treatment indicator. A binary post-opening flag is constructed per treated Costco by merging the hand-collected treatment-date table onto the (Costco × month) panel: 1 from the treatment month forward, 0 before.

Treatment-date validation. Registry first-appearance can lag the true fuel-station opening if a station begins selling fuel before appearing in the state's reporting feed. Each of the four treated Costcos' registry-first-appearance dates was cross-checked against Costco Australia corporate announcements, trade press, and news coverage. The corroborating source must specifically reference the *fuel station* rather than the warehouse; warehouse opening dates are not used because fuel stations and warehouses at Costco Australia regularly open weeks apart. If a fuel-station-specific announced opening date differs from registry first-appearance by more than 30 days, the announced date supersedes.

Costco	**Registry first-appearance**	**Fuel-station-specific corroboration**	**Δ**	**Treatment date used**
Coomera	23 May 2023	23 May 2023 (Retail World; explicit fuel-station opening)	0 d	23 May 2023 (registry)
Casuarina	11 Nov 2022	None; trade press only reports the warehouse opening (7 Dec 2022)	n/a	11 Nov 2022 (registry)
Perth Airport	11 Apr 2020	19 Feb 2020 (ACAPMAg: petrol station selling at $1.18/L; warehouse opens 19 Mar 2020)	−52 d	**19 Feb 2020 (announced)**
Lake Macquarie	2 May 2022	None; no fuel-station-specific opening date in public sources	n/a	2 May 2022 (registry)

Three of the four registry first-appearance dates either match the gas-station-specific corroborated date exactly (Coomera) or lack any gas-station-specific external date for comparison (Casuarina, Lake Macquarie); these retain the registry date. Perth Airport is the only revision: the ACAPMAg report explicitly distinguishes the petrol station (open 19 Feb 2020, with bowser price quoted) from the warehouse (opening one month later on 19 March 2020), placing the fuel-station opening 52 days before the first FuelWatch record.

Station coordinates and distance-to-Costco. Distance is computed as the great-circle distance from each station's coordinates to the nearest Costco fuel station's coordinates. NSW and WA historical registry files do not carry lat/lng, so coordinates are joined from one-time pulls of the live FuelCheck and FuelWatch APIs by (normalized name, postcode); QLD's historical files include coordinates inline. Stations that fail the coordinate join are dropped from analysis (they cannot be classified into a 5 km ring or a donor postcode without a known location); the per-registry join failure rate is reported in (f).

(e) Aggregation

Why monthly. Three considerations point at month as the right level of aggregation. First, NSW and QLD are event-based registries (rows are price-change events) while WA is a daily-snapshot registry; month is the lowest common frequency at which all three speak the same language without having to invent ‘daily’ observations for the event-based feeds. Second, the research question is about sustained competitive response to Costco entry, which plays out over weeks-to-months rather than days, so daily resolution would add noise without identification. Third, monthly means across multiple stations in a ring or postcode smooth single-quote outliers without requiring any winsorisation step.

Two-stage aggregation. Within each station × month, we compute a single mean unleaded price across all of that station's price observations in the month. We then take a simple (unweighted) mean across stations to produce the (ring × month) treated panel and the (postcode × month) donor panel. The two-stage structure prevents stations that change prices more often (and therefore appear more times in the event-based NSW/QLD feeds) from contributing disproportionately to the cell mean.

Carry-forward step (NSW and QLD only). For NSW and QLD, where rows are price-change events rather than daily snapshots, we construct each station's daily price series by carrying its last-observed price forward to subsequent days within the month until the next observed change. This is the standard approach for converting price-event data to a daily series and then to monthly means. WA's daily-snapshot data passes through unchanged.

(f) Data merging

Three sources are joined into a single analysis panel:

- **State registries** (NSW FuelCheck, QLD FPRS, WA FuelWatch) provide the price observations.
- **Live-API station snapshots** (NSW, WA) and inline coordinates (QLD) provide station latitude / longitude, joined to historical registry rows by (normalised station name, postcode).
- **The hand-collected Costco metadata** (`data/catalogs/usable_costcos.csv`) provides each Costco's lat/lng and treatment date, joined to treated panels by Costco name.

Expected losses in the merge. Two sources of leakage. First, stations that closed, rebranded, or were re-numbered before the April 2026 live-API pull are absent from that snapshot, so historical rows that mention them fail the coordinate join. The snapshot misses **2,376 of 4,543 NSW historical stations (52.3%)** and **673 of 1,423 WA historical stations (47.3%)**; QLD is unaffected because its historical files carry coordinates inline. Row-level impact after the upstream fuel-type and sentinel-price filters is reported in (g). Second, station-name typos and mid-panel renames occasionally break the join. We address renames programmatically where we can (Section 1 noted the May 2022 NSW Costco ‘Members only’ suffix; the build script strips it). The loss falls almost entirely on the donor pool (all four treated Costcos join cleanly), and 196 donor postcodes survive the downstream geographic and quality filters.

(g) Filters and transformations: the funnel from raw to analysis-ready

Counts are exact at every step. Row units in the upper block give way to (station × month) cells, (postcode × month) cells, postcode counts, and Costco counts as the panel is aggregated and filtered. Source: `scripts/synthetic_control_input/build_sc_inputs.py`.

Table 2. Sample-construction funnel.

Step	NSW	QLD	WA	Justification
Raw registry pull (rows)	4,401,971	4,970,650	8,047,153	Source data
Filter to regular unleaded (rows)	1,982,507	997,680	2,059,755	Question is about retail unleaded competition; premium/diesel are different markets
Drop sentinel prices (50–500 ¢/L; rows)	1,982,506	995,145	2,059,755	Removes out-of-stock placeholder values; nearly nonbinding once the unleaded and date filters have run
Join station coordinates (rows)	1,247,728	995,145	1,289,912	Required for radius assignment; NSW/WA join from live-API snapshots (per-registry failure rates in (f)); QLD has inline coords
Aggregate to (station × month)	156,864	70,564	68,332	Carry-forward for NSW/QLD events; passthrough for WA snapshots
Aggregate to (postcode × month)	42,307	18,186	18,089	Donor candidate panel (561 / 245 / 198 postcodes) before geographic and quality filters
Drop postcodes ≤20 km from any Costco	−231 pcds	−51 pcds	−80 pcds	Donors must be unambiguously outside any Costco's competitive footprint
Drop postcodes with <3 stations / month avg	−226 pcds	−138 pcds	−82 pcds	Donor monthly mean too noisy below 3 stations
Drop postcodes with <24 months observed	−0 pcds	−0 pcds	−0 pcds	Coverage was good; no postcode failed this gate
Final donor pool	104 pcds	56 pcds	36 pcds	196 postcodes × ~90 months = 17,578 (postcode × month) cells
Costco eligibility (≥24 pre, ≥6 post)	−4 Costcos	−2 Costcos	0	Inclusion rule applied to all 10 Australian Costco fuel stations (5 NSW, 3 QLD, 2 WA)

Final treated panel	1 Costco	1 Costco	2 Costcos	4 Costcos × (5 km ring) × ~80–100 months = 375 (Costco × month) cells

The output is four files in `data/sc_inputs/`: `treated_units.csv` (375 Costco-month rows), `donor_pool.csv` (17,578 postcode-month rows), and two metadata files.

Plan of Attack: Section 3, Analysis Plan

(a) Research design

Question type. Causal. The identification strategy is **synthetic control**, fit separately for each of the four treated Costcos.

Why synthetic control over difference-in-differences. The data has four treated Costcos and 196 candidate donor postcodes. Four treated units is too few to support a credible treated-versus-control mean comparison: a single anomalous Costco (e.g. Casuarina, with only ~3 contributing stations) would dominate the average. Synthetic control instead builds a weighted counterfactual for each treated unit individually, which is exactly the design the n=4 case calls for.

Source of variation. Costco fuel-station entry into a local market isn't driven by what's happening to local fuel prices. Site selection is driven by Costco warehouse logistics, land availability, and Costco's national rollout schedule, none of which respond to the local fuel-price trajectory at the chosen site. The variation we exploit is the staggered timing of the four openings (Feb 2020, May 2022, Nov 2022, May 2023) across three states.

Identifying assumptions. Each assumption is paired with the figure or check that defends it.

- **Pre-period fit captures unobserved time-varying confounders.** Verified visually in Figure 1 (pre-period overlap between treated and synthetic), quantified in Table 2 (pre-period tracking error per Costco), and tested by the 12-month holdout in Robustness Check 5.
- **Donors stay clean.** No donor postcode is partly affected by a Costco that isn't the one we're studying. Defended by the 20 km donor buffer (4× the treated radius), pre-committed in Section 2(b), and explicitly tested by Robustness Check 7 (spatial placebo on the 5–20 km donut).
- **No anticipation.** Competitor stations do not preemptively cut prices ahead of Costco's announced opening. Tested by Robustness Check 1 (placebo in time, −12 months).
- **Range coverage.** The treated ring's pre-period price level can be expressed as a weighted average of donor postcodes. Verified by the donor-weight bars in Figure 2 (no negative weights, no extreme single-donor concentration).

What the weights do. One synthetic-control fit per treated Costco. For each Costco, the donor pool is restricted to *same-state* postcodes only (NSW: 104, QLD: 56, WA: 36, from `data/sc_inputs/donor_pool.csv`) so the synthetic absorbs state-specific shocks (state excise rates, regional refining capacity, intra-state shipping costs) by construction. Weights are chosen to minimise the average gap between the treated 5 km ring's monthly mean unleaded price and the weighted donor mean across the pre-period.

Method commitment. Standard synthetic-control method via the `pysyncon.Synth` library. Predictors: the outcome series (monthly mean unleaded price) only. We have no postcode-level demographic panel that would justify auxiliary covariates. Pre-period objective: minimise the average gap between treated and synthetic on monthly mean unleaded price. Inference: estimate uncertainty by treating each donor postcode as a fake treated unit and measuring the resulting fake effect (Robustness Check 6); summarised as the post/pre tracking-error ratio and as a 95% CI on the post-treatment gap.

(b) Key analyses

Three figures plus two supporting tables. The figures carry the argument; the tables quantify what the figures show. Figures in this document are **illustrative**: they show what we expect each output to look like once the synthetic-control fits are run, not a pre-committed result.

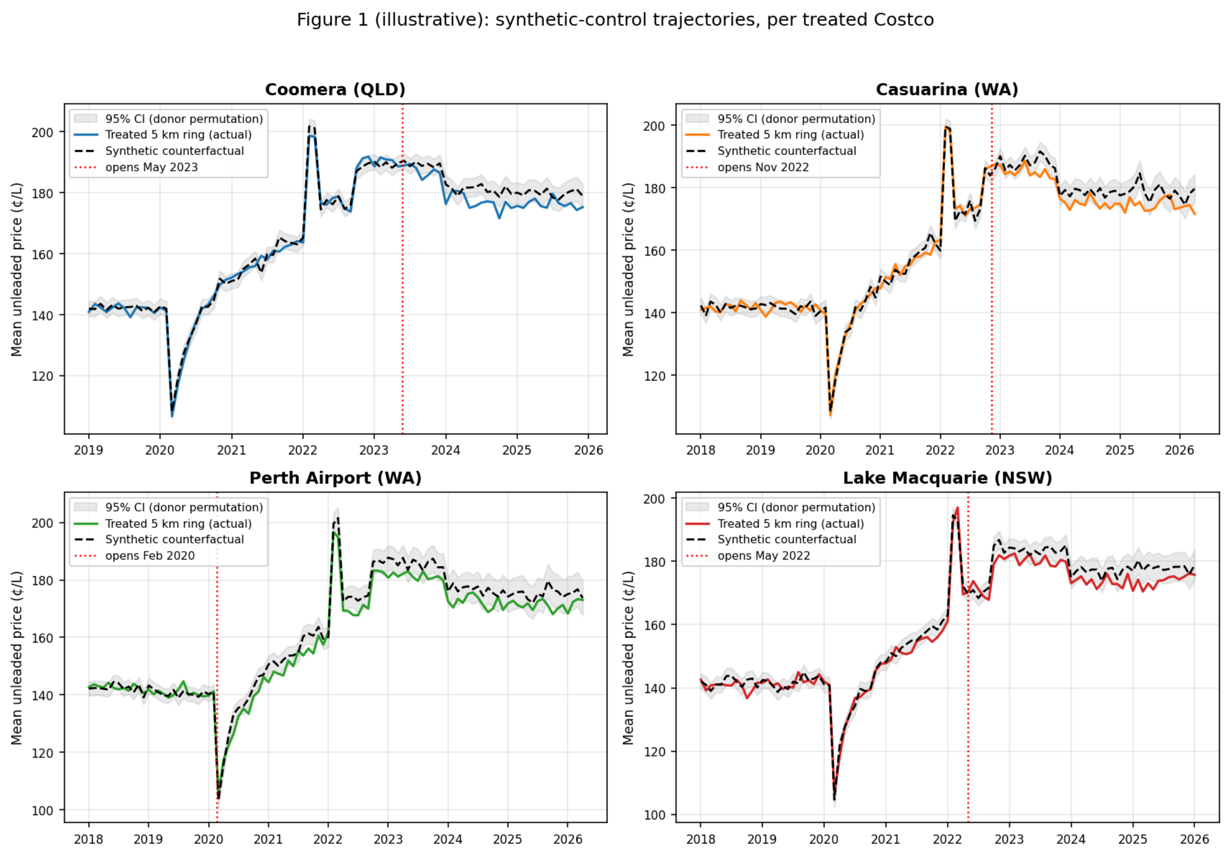
Analysis 1 (Figure 1): synthetic-control trajectories

Test. The headline result. Pre-period overlap between treated and synthetic measures fit quality; the post-period gap is the estimated treatment effect.

Plot. 2×2 panel, one per treated Costco (Coomera, Casuarina, Perth Airport, Lake Macquarie). X-axis: month, from each Costco's pre-period start through the relevant state's registry end (December 2025 for QLD, January 2026 for NSW, April 2026 for WA). Y-axis: monthly mean unleaded price (¢/L). Two lines per panel: solid coloured = treated 5 km ring (actual), dashed black = synthetic counterfactual. Vertical dotted red line at the validated treatment date (Section 2(d)). Shaded grey 95% CI band on the synthetic from donor-permutation inference (Robustness Check 6).

AI-executable instructions. Read `data/sc_inputs/treated_units.csv` and `donor_pool.csv`. For each treated Costco, filter donors to same state, pivot both to wide format (rows = month, columns = unit), and fit `pysyncon.Synth` on pre-period only (months strictly before the treatment date). Project the synthetic across the full panel. Build the 95% CI by re-fitting synthetic-control on each donor as a placebo and taking the empirical 2.5–97.5 percentile of donor-side gaps at each post-treatment month. Render to `section_3/plots/01_sc_trajectories.png`.

Why this analysis. A reader who only sees Figure 1 should be able to read off the project's conclusion: did competitor prices fall after Costco entry, and by how much. It is the canonical synthetic-control visualisation.



Illustrative Figure 1 (hypothetical data). Solid coloured = treated 5 km ring; dashed black = synthetic counterfactual; vertical dotted red = validated Costco opening date; shaded grey = 95% donor-permutation CI on the synthetic. The illustrative gap of ~4 ¢/L emerges within a few months of treatment and persists; the actual fits will reveal whether and by how much the real data exhibits this pattern.

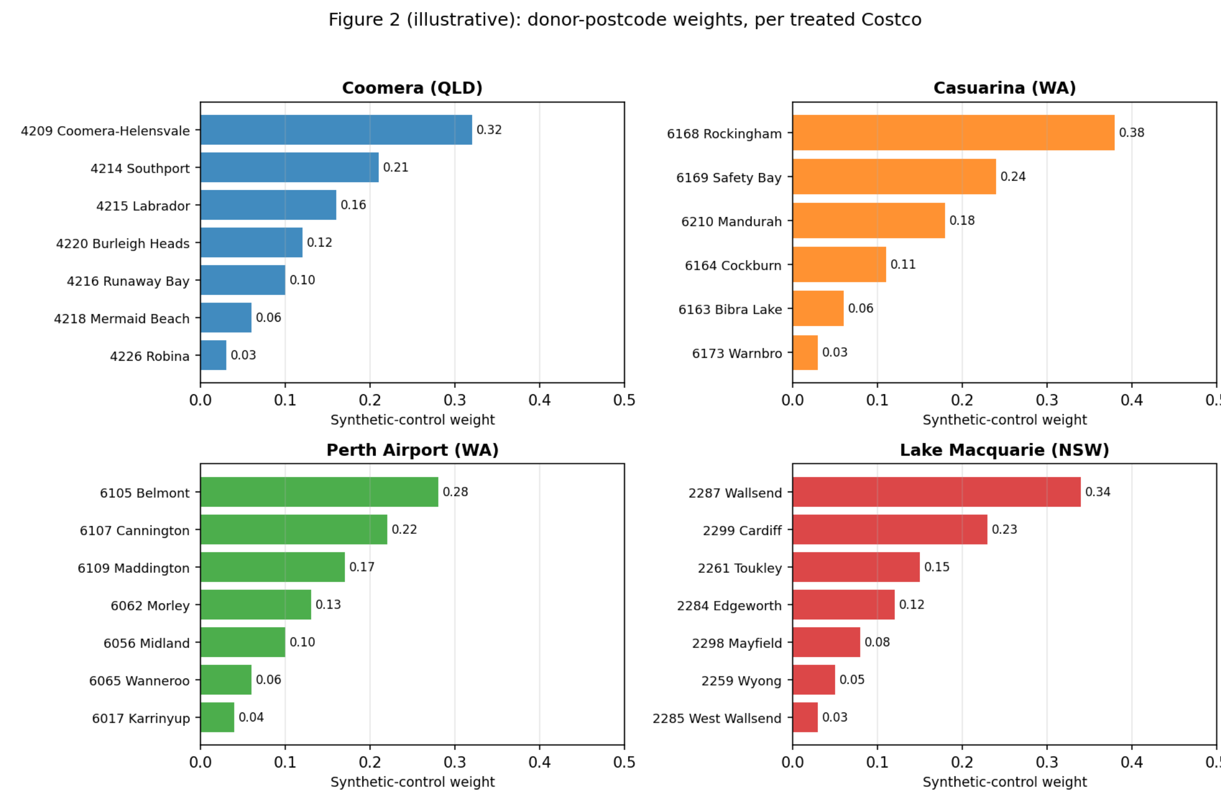
Analysis 2 (Figure 2): donor weights per treated Costco

Test. Transparency about which donors drive each counterfactual; common-support assumption.

Plot. 2×2 panel of horizontal bar charts, one per treated Costco. One bar per donor postcode with weight > 0.01, sorted descending, annotated with postcode and suburb (suburb joined from `donor_metadata.csv`).

AI-executable instructions. Extract the weight vector from each fitted SC model in Analysis 1. Filter to weights > 0.01. Sort descending. Render to `section_3/plots/02_donor_weights.png`.

Why this analysis. A counterfactual leaning on one or two donors is fragile; one drawn from a balanced mix of geographically sensible donors is credible. The per-Costco panels also let the reader sanity-check geographic plausibility. Coomera should lean on Gold Coast and Brisbane postcodes, not far-North Queensland; Lake Macquarie should lean on Newcastle and the Central Coast. This figure also flags any common-support concerns.



Illustrative Figure 2 (hypothetical data). Synthetic-control weights for the top donor postcodes per treated Costco, sorted descending. The illustrative weight distributions are deliberately spread across 6–7 donors with no single donor exceeding 0.40, reflecting a credible synthetic; weights concentrated in 1–2 donors would be a flag for a fragile counterfactual.

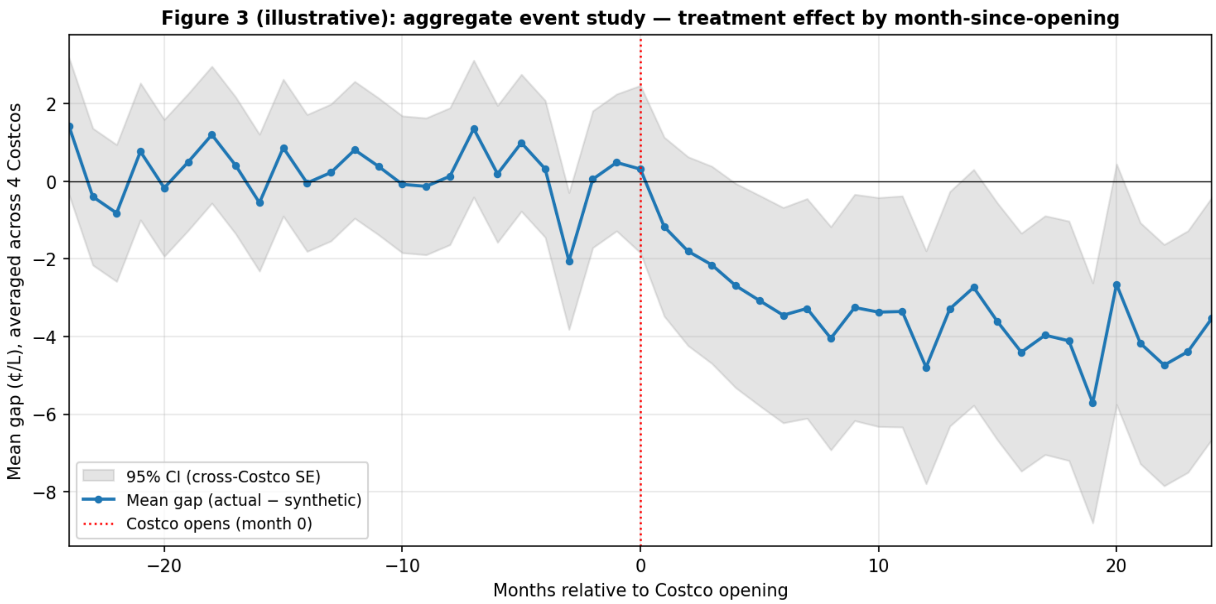
Analysis 3 (Figure 3): aggregate event study

Test. The dynamic profile of the effect. Do competitors respond immediately, with delay, persistently, or with a fading effect?

Plot. Single-panel line plot. X-axis: months relative to Costco opening, –24 to +24. Y-axis: mean (actual – synthetic) gap in ¢/L, averaged across the four treated Costcos. Shaded grey 95% CI band from cross-Costco standard error. Vertical dotted red line at month 0 (treatment).

AI-executable instructions. Take per-Costco gap series (actual – synthetic) from Analysis 1. Re-index each to relative-month-since-treatment. At each relative month in [–24, +24], average the four Costco gaps and compute the cross-Costco SE for the band. Render to `section_3/plots/03_event_study_aggregate.png`.

Why this analysis. The 2×2 panel in Figure 1 shows the per-Costco answer but obscures the aggregate dynamic. Figure 3 turns ‘Costco lowered prices on average’ into ‘Costco lowered prices by X ¢/L within Y months and the effect persisted for Z’, much closer to a managerial answer. The –24 to +24 window is the largest range over which all four Costcos contribute (limited by Coomera's 31 post-months and Perth Airport's 26 pre-months).



Illustrative Figure 3 (hypothetical data). Mean gap (actual – synthetic) by month relative to Costco opening, averaged across the four treated Costcos. The illustrative profile shows a pre-period gap fluctuating around zero (the identifying assumption that synthetic tracks treated absent the intervention) and a post-period gap that opens smoothly over ~4 months and stabilises near –4 ¢/L. The actual event study will reveal whether the real effect is immediate, delayed, persistent, or fading.

Analysis 4: supporting tables

Two tables that quantify what Figures 1 and 3 show. Both are computed alongside Analysis 1 from the fitted SC models and the permutation distribution.

Table 1. Per-Costco effect-size summary.

Costco	Pre-period mean (¢/L)	Mean post-treatment gap (¢/L)	% reduction	Post/pre tracking-error ratio	95% CI on gap	Post months
Coomera	—	—	—	—	—	—
Casuarina	—	—	—	—	—	—
Perth Airport	—	—	—	—	—	—
Lake Macquarie	—	—	—	—	—	—

Table values populate from the fitted SC models. The post/pre tracking-error ratio compares how far the synthetic strays from actual after Costco entry to how closely it tracked before. A ratio of 4–5 or higher is the standard threshold for evidence of a treatment effect.

Table 2. Pre-period fit quality.

Costco	Pre months	Pre tracking error (¢/L)	Tracking error as % of pre mean	Top-3 donor weight share
Coomera	51	—	—	—
Casuarina	58	—	—	—

Perth Airport	26	—	—	—
Lake Macquarie	58	—	—	—

Table values populate from the fitted SC models. Lets the reader judge per-unit identification credibility at a glance. Casuarina's pre-period tracking error is expected to be visibly worse than Perth Airport's because the 5 km Casuarina ring contains only ~3 contributing stations on average (Section 1's documented limitation); surfacing it transparently here is part of presenting that limitation rather than hiding it.

(c) Robustness checks (extra credit)

The biggest sources of fragility in the headline analysis are (i) the 5 km / 20 km geographic geometry, (ii) the COVID period coinciding with Perth Airport's opening, (iii) the noise in Casuarina's sparse 5 km ring, (iv) the standard synthetic-control concern that pre-period fit can be gamed by over-flexible donor selection, and (v) the lack of formal uncertainty quantification on a four-Costco point estimate. The seven checks below directly probe these concerns.

Table 3. Robustness check menu.

#	Check	Concern it tests	Pass criterion
1	Placebo in time (–12 months)	No-anticipation assumption	Placebo gap near zero, within donor-side 95% CI
2	Alternative radii (3 / 8 km treated; 15 / 30 km donor)	Geographic geometry choice	Effect sign and approximate magnitude survive
3	Casuarina at 10 km treated radius	Sparse-ring noise (~3 stations at 5 km)	10 km gap comparable in sign and magnitude to 5 km
4	Perth Airport with March–December 2020 dropped	COVID demand collapse contaminates early post-period	Post-COVID gap comparable to full-window gap
5	LOOCV donor selection + 12-month holdout	Donor-selection overfitting	Holdout tracking error comparable to in-sample training error
6	95% CIs via permutation inference	Statistical uncertainty on the gap	CI excludes zero for high-coverage Costcos
7	Spatial placebo: 5–20 km donut as faux donors	Donor purity / extent of Costco's spatial footprint	Either result informative (see paragraph below)

Check 1: placebo in time. For each treated Costco, backdate treatment by 12 months. Re-fit synthetic control on data up to the placebo date; project forward; compute the placebo gap. Drop all real-treatment-and-after months from the placebo fit so no real post-treatment data contaminates the placebo window. *Concern:* if a ‘ban-style effect’ appears in the placebo window before Costco actually opens, the post-treatment gap reflects pre-existing trends or anticipation, not Costco entry.

Check 2: alternative radii. Re-build synthetic-control inputs at (3 km treated, 15 km donor) and (8 km treated, 30 km donor). Re-fit Analysis 1 for each Costco at each geometry. Report the resulting gaps in a sensitivity table. *Concern:* the 5 km / 20 km baseline geometry is defensible (see Section 2(b)) but discretionary; a result that exists at 5 km but vanishes at 3 km or 8 km would be fragile. This is the project's most important sensitivity test because the radius geometry *is* the design.

Check 3: Casuarina at 10 km. Casuarina's 5 km ring contains only ~3 contributing stations on average, the documented limitation in Section 1. A wider ring captures more signal at the cost of including some less-exposed stations. Re-fit Casuarina specifically at a 10 km treated radius (donor buffer remains 20 km). Compare gap and pre-period tracking error to the 5 km baseline. *Concern:* the headline Casuarina estimate may be driven by individual-station noise rather than competitive response.

Check 4: Perth Airport with COVID-period removal. Perth Airport opens approximately four weeks before Australia's COVID demand collapse begins. Early post-treatment months are dominated by pandemic-driven price movements rather than competitive response. Re-fit Perth Airport with March–December 2020 dropped from both treated and donor panels. *Concern:* COVID compression of fuel prices may suppress the actual − synthetic gap; the corrected estimate may be larger in magnitude than the full-window gap.

Check 5: LOOCV donor selection with 12-month holdout. Pre-period fit can be gamed by greedy donor selection (cherry-picking donors that fit the pre-period without generalising). The check has four steps. (1) Split each Costco's pre-treatment period into a *training window* (all pre-months except the last 12) and a *12-month holdout*. The holdout is 12 months rather than 24 because Perth Airport has only 26 pre-months; a 24-month holdout would leave too few training months to fit weights. (2) For each candidate N donors ∈ {5, 10, 20, 50, full}: leave-one-out cross-validation within the training window. Hold out one training month at a time, refit weights on the rest, score tracking error at the held-out month; pick the N that minimises mean LOOCV error. (3) Refit the synthetic with the chosen N on the full training window. (4) *Without looking at any post-treatment data*, check whether the synthetic continues to track the treated series across the untouched 12-month holdout. *Concern:* over-flexible donor selection can produce a tight pre-period fit that does not generalise; the holdout is the honest proof that the synthetic was not memorising the training series.

Check 6: 95% CIs via permutation inference. Standard synthetic-control permutation. For each donor postcode, treat it as a ‘placebo treated’ and fit synthetic control using the remaining donors. Compute the placebo gap series. Across all donor placebos, take the empirical 2.5–97.5 percentile of post-treatment gaps to construct a 95% CI on the real treatment effect. Reported on Figure 1 as a shaded band and in Table 1 as a column. *Concern:* point estimates without uncertainty bands can make a modest effect look airtight. For Casuarina specifically, a CI that includes zero will be reported transparently as the documented limitation, consistent with Section 1's noted noise issue.

Check 7: spatial placebo (5–20 km donut as faux donors). The 20 km donor buffer is meant to keep donors fully outside any Costco's competitive zone, but this assumption is never verified in the main analysis. The 5–20 km donut stations (currently excluded from both treated and donor groups) are a natural laboratory: treat them as a faux donor pool, fit synthetic control, see whether they show a ‘Costco effect’ post-opening. Two outcomes are both informative: (a) the donut shows an effect, meaning Costco's footprint extends beyond 5 km and our headline estimate is conservative (the true effect is probably larger); (b) the donut shows no effect, meaning the Costco effect is concentrated within 5 km and the 20 km buffer is more than sufficient. *Concern:* the spatial analog of the temporal placebo. Whichever direction the result lands, it tightens the interpretation of the headline number.